Quiz: Discrete and Continuous Random Variables (Application-Based)

Time Allowed: 15 Minutes
Total Marks: 10

Instructions: Attempt all questions. Show necessary steps.

Q1. A shop records the number of customers X arriving in a short interval. The probabilities are given by:

$$P(X = x) = \begin{cases} k, & x = 0 \\ 2k, & x = 1 \\ 3k, & x = 2 \\ 4k, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of k.

(b) Find the probability that at least 2 customers arrive.

(c) Find the expected number of customers.

Q2. A machine produces items labeled $1, 2, 3, 4, 5$. The probability of producing item x is proportional to $(6 - x)$.

(a) Find the probability mass function.

(b) Find the probability that the item number is even.

(c) Find $\text{Var}(X)$.

Q3. The lifetime (in hours) of a component is modeled by a continuous random variable X with density:

$$f(x) = \begin{cases} c(2 - x), & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find c.

(b) Find the probability that the lifetime exceeds 1 hour.

(c) Find the mean lifetime.

Q4. The cumulative distribution function of the waiting time X (in minutes) is:

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{4}, & 0 \le x \le 2 \\ 1, & x > 2 \end{cases}$$

(a) Find the probability density function.

(b) Find the probability that waiting time is between 1 and 2 minutes.

(c) Find the median waiting time.

Q5. The amount of water (in liters) used by a household per day is modeled by:

$$f(x) = \begin{cases} kx^2, & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find k.

(b) Find the probability that usage exceeds 1 liter.

(c) Find $E(X)$.